

Git & GitHub



Git

Version Control System is a tool that helps to track changes in code

Git is a **Version Control System**. It is :

- *popular*
- *free& Open Source*
- *fast & scalable*



GitHub

Website that allows developers to store and manage their code using Git.

<https://github.com>



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GitHub Account

- Create a new repository : alphaknowledge-demo
- Make our first commit



Setting up Git

Visual Studio Code

Windows (Git Bash)

Mac(Terminal)

```
git --version
```



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Configuring Git

```
git config --global user.name "Alpha Knowledge"
```

```
git config --global user.email "alpha@gmail.com"
```

```
git config --list
```



Clone & Status

Clone - Cloning a repository on our local machine

```
git clone <-some link->
```

Status - displays the state of the code

```
git status
```



untracked - new files that git doesn't yet track

modified - changed

staged- file is ready to be committed

unmodified - unchanged



Add & Commit

add- adds new or changed file in your working directory to the Git staging area.

git add <-file name->

commit- it is the record of change

git commit -m "some message"



Push Command

push- upload local repo content to remote repo

git push origin main



Init Command

init - used to create a new git repo

git init

git remote add origin <-link->

git remote -v (to verify remote)

git branch (to check branch)

git branch -M main (to rename branch)

git push origin main



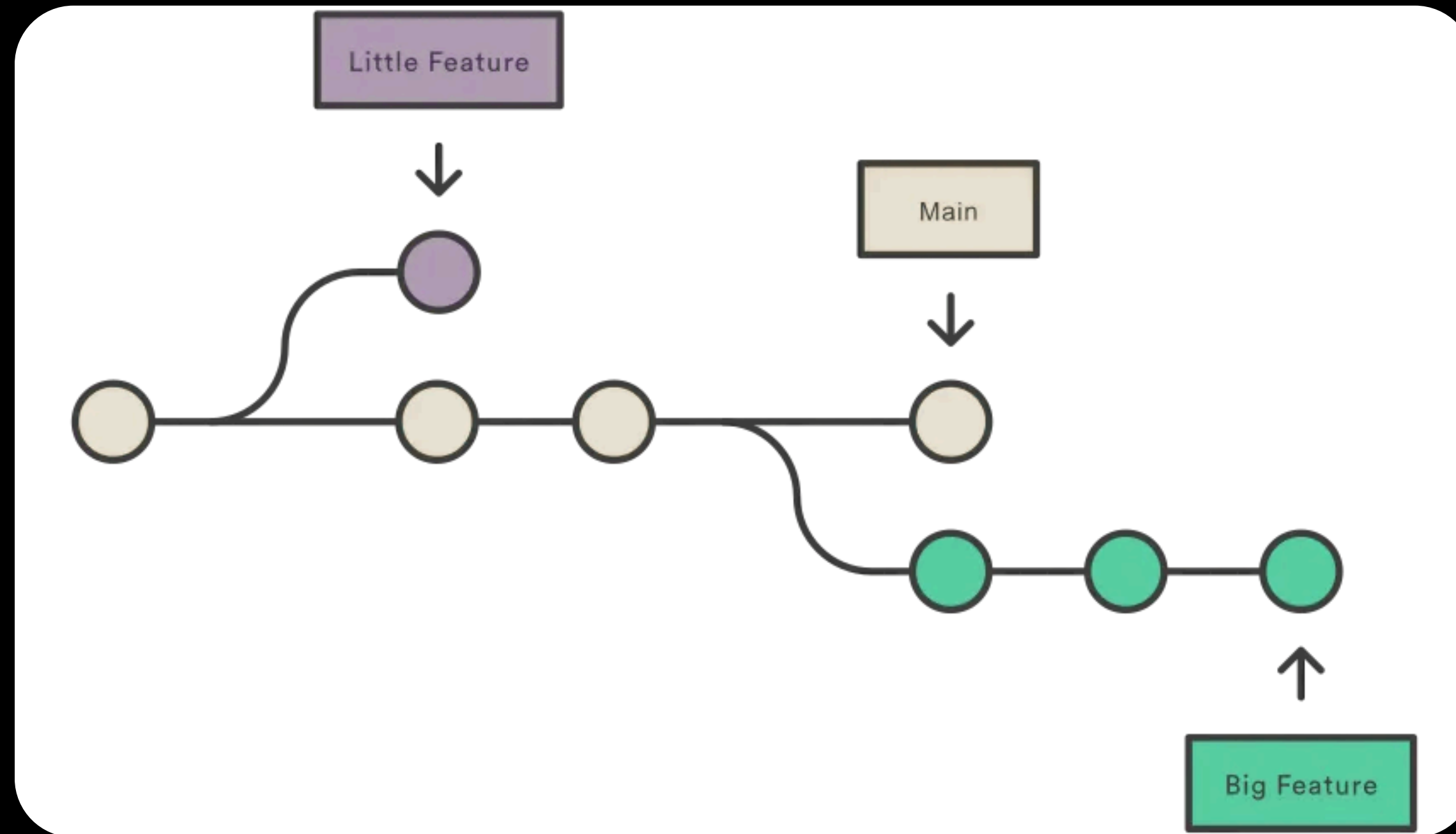
Work Flow

Local Git



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Git Branches



Branch Commands

git branch (to check branch)

git branch -M main (to rename branch)

git checkout <-branch name-> (to navigate)

git checkout -b <-new branch name-> (to create new branch)

git branch -d <-branch name-> (to delete branch)



Merging Code

Way 1

git diff <-branch name-> (to compare commits, branches, files & more)

git merge <-branch name-> (to merge 2 branches)

Way 2

Create a PR



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Pull Request

It lets you tell others about changes you've pushed to a branch in a repository on GitHub.



Pull Command

```
git pull origin main
```

used to fetch and download content from a remote repo and immediately update the local repo to match the content



Resolving Merge Conflicts

An event that takes place when a Git is unable to automatically resolve differences in code between two commits.



Undoing Changes

Case 1 : staged changes

```
git reset <-file name->
```

```
git reset
```

Case 2: committed changes (for one commit)

```
git reset HEAD-1
```

Case 3: committed changes (for many commits)

```
git reset <-commit hash->
```

```
git reset --hard <-commit hash->
```



Fork

A fork is a new repository that shares code and visibility settings with the original “**upstream**” repository.

Fork is a rough copy

